Audit Data Dictionary

analysis_confidence

Deterministic confidence label derived from analysis_reason_code. This is not the same as research_confidence. It reflects the pipeline's confidence in the current analysis classification.

Possible values, ordered by confidence:

- Blank = no analysis reason code is assigned.
- LOW = lower deterministic confidence, usually unresolved or less isolated diagnostics.
- MEDIUM = medium deterministic confidence, including source1-focused diagnostics such as missing event adjustment, adjustment-factor continuity, partial/extra event capture, high-score anomaly, and methodology/source diagnostics such as event denominator or event source mismatch.
- HIGH = high deterministic confidence, such as close reversal, source2 dividend/split factor mismatch, or source2 event-date mismatch.

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analysis_reason_code

Deterministic diagnostic classification assigned by the audit pipeline. It explains the main reason the row needs review. When real-world research is available, the code may be adjusted so the diagnostic agrees with likely_correct_source.

Possible values:

- CLOSE_REVERSAL = source1/source2 return difference reverses on an adjacent trading day, suggesting a close-source or timing artifact rather than a corporate action.
  Example: On Tuesday, source1 return is -2.00% and source2 return is -3.00%, so diff_return is +1.00%. On Wednesday, source1 return is +1.00% and source2 return is +2.00%, so diff_return is -1.00%. The equal and opposite differences suggest a close timing issue, not a real event.
- EVENT_DENOMINATOR_MISMATCH = source1 and source2 record the same dividend/split marker, but the event-return percentage differs because the same cash amount is divided by different prior-close values.
  Example: source1 reports cd:0.52, source2 reports ca:0.52, and both sources show the same raw price return, but source1's prior close is $28.00 while source2's prior close is $21.00. The same $0.52 cash dividend therefore produces different event-return percentages.
- EVENT_SOURCE_MISMATCH = source1 and source2 report different dividend/split event markers for the date, especially when both sources have event markers but use different event formatting, grouping, amounts, or source-event representation. When both sources

have same-day event markers and the residual return difference is small, this is usually an event-representation issue; it can also be used as a broader fallback when event markers differ and pre-research evidence does not determine which source is economically correct.

Example: On the same ticker/date, source1 reports cd:0.25 while source2 reports ca:0.50. With a $50.00 prior close, that is about +0.50% versus +1.00% expected dividend impact.

- HIGH_SCORE_ANOMALY = source1 return has a high heuristic anomaly score even when the source1/source2 return difference is not material.

  Example: source1 and source2 both show +15.00% for the day, so diff_return is 0.00%. The ticker's recent daily moves are usually near 1.00%, so the +15.00% move still needs review.
- SOURCE1_ADJ_FACTOR_CONTINUITY = Rare fallback diagnostic for an unexplained source1 adjustment-chain discontinuity. This situation is expected to be uncommon and may not appear in ordinary audit runs; it exists as a guardrail for cases where source1's cumulative adjustment-factor change does not align with the adjusted-return difference and the discrepancy is not better explained by a same-day event-source, event-denominator, missing-event, partial-event, or extra-event diagnosis.

  Example: Both sources show close rising from $100.00 to $101.00, a +1.00% price return. Both also show a $1.00 dividend, so the adjusted return should be about +2.00%. source1's adjusted-return chain shows +1.20%, and no more specific event or denominator issue explains the difference.
- SOURCE1_EVENT_DATE_MISMATCH = source1 appears to have an event on the wrong trading date.

  Example: A $0.50 dividend has confirmed ex-date Tuesday. source1 shows cd:0.50 on Monday and no event on Tuesday, while source2 shows ca:0.50 on Tuesday.
- SOURCE1_EXTRA_EVENT = source1 records an extra corporate-action event or extra event amount not supported by source2 or by the adjusted-return difference.

  Example: Research confirms one $0.65 dividend. source1 records cd:0.65 cd:0.65, source2 records ca:0.65, and diff_return reconciles to source1's extra $0.65 event-return impact.
- SOURCE1_MISSING_EVENT = source1 appears to be missing a corporate-action event or related adjustment needed to explain the return difference.

  Example: source2 records sp:1.05 for a confirmed spin-off, implying about +5.00% adjusted-return impact. source1 has no event marker and shows a return about 5.00 percentage points lower.
- SOURCE1_PARTIAL_EVENT = source1 records a same-day corporate-action event, but the recorded event amount appears incomplete relative to the source2 event record and the adjusted-return difference.

  Example: Research confirms a $0.15 dividend made up of a $0.075 base dividend plus a $0.075 variable dividend. source1 records cd:0.075, source2 records ca:0.15, and diff_return reconciles to the missing $0.075 event-return impact.

- SOURCE1_RETURN_METHOD_UNRESOLVED = Rare final fallback diagnostic for a source1/source2 return difference that remains unexplained after the specific event, denominator, date, close-reversal, adjustment-continuity, and dividend/split reconciliation checks have been applied. This situation is unexpected in ordinary audit runs; it exists as a guardrail for cases where the input fields do not isolate a single deterministic cause.
  Example: source1 return is +2.00% and source2 return is +3.20%, so diff_return is -1.20%. There is no event marker mismatch, denominator mismatch, factor-continuity mismatch, dividend/split reconciliation break, or adjacent-day reversal to explain it.
- SOURCE2_DIV_SPLIT_RETURN_MISMATCH = source2 implied dividend/split factor materially does not reconcile to source2 explicit dividend/split factor after allowing for ordinary cash-dividend convention differences.
  Example: source2 records a $0.40 dividend and prior close is $40.00, so the explicit dividend impact is about +1.00%. source2's adjusted close implies a +2.50% dividend/split impact instead.
- SOURCE2_EVENT_DATE_MISMATCH = post-research override indicating source2 appears to have the event on the wrong trading date.
  Example: Research confirms a $0.60 dividend belongs on Thursday. source1 shows cd:0.60 on Thursday, but source2 shows ca:0.60 on Wednesday.
- SOURCE2_MISSING_EVENT = post-research override indicating source2 appears to be missing a corporate-action event or related adjustment.
  Example: Research confirms a $0.75 dividend with prior close $75.00, so the expected impact is about +1.00%. source1 shows cd:0.75, but source2 has no dividend marker or adjusted-return impact.

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date

Trading date being reconciled. This is the date of the source1 row and the date used to join source2 prices, dividend/split event markers, return calculations, diagnostics, and optional real-world event research.

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diff_return

Material adjusted-return difference between source1 and source2, using the sign convention source1_return - source2_return. Positive means source1's adjusted return is higher than source2's; negative means source1's adjusted return is lower. Differences below the configured 1e-4 tolerance are set to null.

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event_bucket

External research classification for the identified real-world explanation.

Possible values:
- DISTRIBUTION = cash dividend, special dividend, return of capital, or similar cash/non-split distribution.
- MERGER = merger, acquisition, exchange offer, or transaction consideration affecting the price/return series.
- NEWS = material company, industry, macro, earnings, guidance, regulatory, litigation, or analyst/news event rather than a mechanical corporate action.
- PRICING_METHOD = a non-corporate-action difference caused by different data source price, adjustment, or timing methods, after corporate actions and material news have been investigated.
- RIGHTS = rights offering, warrant distribution, subscription right, or similar shareholder right.
- SPINOFF = spin-off, split-off, or separation distribution.
- SPLIT = stock split, reverse split, or stock dividend treated as a split factor.
- UNKNOWN = no sufficiently supported real-world explanation was identified.

The analyst instructions require event precedence in this order: SPLIT, SPINOFF, DISTRIBUTION, MERGER, RIGHTS, NEWS, PRICING_METHOD, UNKNOWN.

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event_detected

External research field indicating whether a real-world event was identified for the ticker/date.

Possible values:
- Blank = no external real-world event research was joined.
- NO = research did not identify a relevant real-world event.
- UNCERTAIN = research did not support a firm yes/no conclusion.
- YES = research identified a relevant real-world event.

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evidence_summary

External research narrative summary. Per the analyst instructions, it should be row-specific, name source1 and source2 explicitly, summarize sources reviewed, describe the real-world event or absence of one, explain economic plausibility, and tie the conclusion back to input row math such as source1_return, source2_return, diff_return, event markers, and expected return impact.

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expected_return_impact

External research estimate of the event's expected return impact using the audit's event-return convention. For splits, the instructions define this as split_factor - 1.0. For cash distributions, the approximate convention is cash_amount / prior_close. This value is used by real_world_events.apply_reason_overrides() to test whether the real-world event magnitude reconciles to signed diff_return within tolerance.

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has_div_split_mismatch

Boolean flag indicating whether source1 and source2 report different dividend/split marker strings for the ticker/date. The comparison preserves source marker text in the output, but normalizes equivalent cash-action prefixes only while comparing: source1 cd and sc markers compare as generic ca cash-action markers against source2.

Possible values:

- false = source1 and source2 dividend/split marker strings match after normalization.
- true = source1 and source2 dividend/split marker strings differ after normalization.

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heuristic_anomaly_score

Deterministic pre-research anomaly score generated from source1-side return behavior. The score is not a probability and does not identify the correct data source by itself. It is a routing signal that helps identify rows where source1's adjusted return looks unusual, even when source1 and source2 returns agree.

Higher scores indicate that more independent anomaly signals were present.

Typical interpretation:

- 0 = no independent source1-side anomaly signals were triggered. The row may still require review because of a data source return difference, event-marker mismatch, or other reconciliation issue.
- 1-3 = low anomaly signal. One or more mild conditions may be present, but the row is usually reviewed only if another diagnostic also requires attention.
- 4-6 = moderate anomaly signal. The source1 return may be unusually large, inconsistent with nearby price behavior, or close to a corporate-action-style return pattern. These rows are worth review, especially when paired with event or adjustment differences.
- 7-10 = high anomaly signal. Multiple independent checks suggest the source1 return is unusual relative to price movement, adjustment mechanics, nearby returns, or corporate-action expectations.
- 11+ = very high anomaly signal. The row has several strong anomaly indicators and should be reviewed even if source1 and source2 returns are identical, because both data sources may be reflecting a real market event or both may share a questionable treatment.

Important notes:

- The score is additive: multiple smaller signals can produce a high score.
- The score is pre-research only. Real-world research and input row math determine the final classification.
- A high score does not imply source1 is wrong. It means the row deserves investigation.
- A score of 0 does not imply the row is correct. It may still be flagged by return differences, event mismatches, adjustment-factor issues, or close-reversal diagnostics.

likely_correct_source

External research conclusion identifying which source appears economically correct after considering both real-world evidence and input row math. The analyst instructions require this to follow from source1_return, source2_return, diff_return, event markers, event-return fields, adjustment factors, and external evidence.

Possible values:

- Blank = no external real-world event research was joined.
- BOTH = source1 and source2 are economically equivalent or both appear reasonable for the row.
- SOURCE1 = source1 appears economically correct.
- NEITHER = neither source1 nor source2 appears economically correct.
- UNCERTAIN = research cannot determine the likely correct source.
- SOURCE2 = source2 appears economically correct.

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needs_review

Boolean review flag. True when diff_return is non-null or when the source1 heuristic anomaly score is greater than or equal to MIN_SCORE_TO_REVIEW, which is currently 5.

Possible values:

- false = row does not meet review criteria.
- true = row meets review criteria.

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primary_url

Primary URL supporting the real-world event research conclusion. The instructions prefer authoritative sources such as SEC filings, investor relations releases, exchange notices, or official corporate-action announcements where available.

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real_world_event

External research description of the real-world event or explanation associated with the ticker/date. This may describe a split, distribution, merger, rights offering, news event, pricing-method explanation, or the absence/uncertainty of a real event depending on the analyst output.

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real_world_evidence

Summary-report-only display column that vertically combines real_world_event and evidence_summary with a blank line between the two sections. It gives the event headline followed by the supporting research and input row reconciliation.

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research_confidence

External research confidence label for the likely_correct_source and event conclusion. This is separate from analysis_confidence: research_confidence comes from the analyst/research evidence scoring rules, while analysis_confidence comes from the deterministic audit classification.

Possible values, ordered by confidence:
- Blank = no external real-world event research was joined.
- LOW = weak, ambiguous, or absence-of-event support.
- MEDIUM = reasonable but incomplete or partly inferential support.
- HIGH = strong source support and clear return/event reconciliation.

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review_priority

Review priority bucket. This is not the same as source1_fix_priority: review_priority records why the row merited review, while source1_fix_priority records whether source1 has a recommended fix after research.

Possible values, ordered by priority:
- Blank = no review is required.
- LOW = heuristic_anomaly_score is at least MIN_SCORE_TO_REVIEW but higher-priority conditions do not apply.
- MEDIUM = heuristic_anomaly_score is at least 8 and the row may be actionable.
- HIGH = diff_return is non-null.

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secondary_url

Secondary or confirming URL supporting the real-world event research conclusion. The instructions allow credible independent sources such as Reuters, Bloomberg, Yahoo Finance, Nasdaq, MarketWatch, Macrotrends, or company press releases.

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source1_adj_close

source1 adjusted close calculated by this audit pipeline as source1_close * source1_adj_factor. This is the source1-side adjusted close used to calculate source1_return.

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source1_adj_factor

source1 cumulative backward-looking adjustment factor calculated by this audit pipeline from source1 split and dividend records. For splits, the factor uses split_from / split_to. For dividends, the factor uses (prior_close - cash_amount) / prior_close. The cumulative factor includes events after the row date so historical closes can be restated on a current adjusted basis.

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source1_close

source1 unadjusted close price for the ticker/date. This comes from source1 unadjusted OHLCV data and is used as the raw close input for source1 adjusted-close and raw price-return calculations.

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source1_div_split

Compact source1 corporate-action marker string for the ticker/date. Split events are represented as sp:<split_factor>, where split_factor is split_to / split_from. Regular cash dividends are represented as cd:<cash_amount>. Special cash dividends are represented as sc:<cash_amount>. Generic cash distributions with no source1 regular/special distinction are represented as ca:<cash_amount>. Multiple same-day events are space-separated.

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source1_fix_action

Suggested source1 review or remediation action, generated from analysis_reason_code and optional real-world research. When research concludes likely_correct_source is SOURCE1 or BOTH, this is cleared because no source1 remediation is recommended. Otherwise examples include adding or correcting missing corporate actions, reviewing adjustment-factor history or event denominators, rebuilding adjusted close and return chains, or manually reviewing close/adjusted-close/corporate-action inputs.

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source1_fix_priority

Priority for the suggested source1 remediation. This is research-aware: it is blank when real-world research concludes likely_correct_source is SOURCE1 or BOTH.

Possible values, ordered by priority:

- Blank = no source1 remediation priority is assigned.
- LOW = low-priority source1 review, usually lower-score heuristic anomaly.
- MEDIUM = more investigative source1 review, such as an unresolved adjusted-return calculation issue or higher-score anomaly.
- HIGH = direct source1 fix candidate, such as missing event adjustment, event-date mismatch, adjustment-factor continuity, partial event capture, or extra event capture.

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source1_needs_fix

Boolean flag indicating whether source1 may require correction after deterministic diagnostics and optional real-world research. If research concludes likely_correct_source is SOURCE1 or BOTH, this is false. If research concludes likely_correct_source is SOURCE2 or NEITHER, this is true. Without research, it follows source1-focused reason codes such as missing event adjustment, event-date mismatch, adjustment-factor continuity issue, partial event capture, extra event capture, unresolved source1 adjusted-return calculation issue, or high-score anomaly. Generic EVENT_SOURCE_MISMATCH rows require review but do not by themselves imply source1 needs a fix.

Possible values:

- false = source1 does not currently appear to need correction.
- true = source1 may require correction or manual remediation review.

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source1_problem_and_fix

Summary-report-only display column that vertically combines source1_problem_summary, source1_why_incorrect, and source1_fix_action with blank lines between the three sections.

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source1_problem_summary

Human-readable summary of the audit issue or suspected source1-side problem, generated from analysis_reason_code and optional real-world research. Blank when the pipeline does not identify a review issue or when research concludes likely_correct_source is SOURCE1 or BOTH.

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source1_return

Final source1 adjusted return for the ticker/date, calculated from the percentage change in the pipeline's source1 adjusted close series.

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source1_return_price

source1 raw price return from unadjusted closes, calculated as the percentage change from the prior source1 unadjusted close to the current source1 unadjusted close for the same ticker.

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source1_why_incorrect

Human-readable explanation of why source1 may be incorrect or why the row needs source1 review/context, generated from analysis_reason_code and optional real-world research. Blank when the pipeline does not identify a review issue or when research concludes likely_correct_source is SOURCE1 or BOTH.

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source2_adj_close

source2 adjusted close for the ticker/date, taken from source2 OHLCV data. This is the adjusted close used to calculate source2_return.

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source2_adj_factor

Implied source2 adjustment factor for the ticker/date, calculated as source2_adj_close / source2_close when source2_close is present and nonzero. Unlike source1_adj_factor, this is inferred from source2's supplied adjusted close rather than rebuilt directly from source2 event records.

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source2_close

source2 unadjusted close price for the ticker/date. This comes from source2 OHLCV data and is used for source2 raw price-return calculations and to infer the source2 adjustment factor.

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source2_div_split

Compact source2 corporate-action marker string for the ticker/date. Split events are represented as sp:<split_factor>, where split_factor is source2's split_ratio. Cash dividend/distribution events are represented as ca:<cash_amount> because source2 does not expose the same source1 regular cash dividend cd versus special cash dividend sc distinction. Multiple same-day events are space-separated.

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source2_return

Final source2 adjusted return for the ticker/date, calculated from the percentage change in source2 adjusted close.

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source2_return_price

source2 raw price return from unadjusted closes, calculated as the percentage change from the prior source2 unadjusted close to the current source2 unadjusted close for the same ticker.

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ticker

Normalized ticker symbol being audited. The pipeline strips surrounding whitespace and uppercases ticker symbols before joining source1 and source2 data.
